

Lecture 7

Fitting a mixture of Gaussians

COURSE MATH 637: Mathematical Techniques in Data Science

WEBSITE <https://jacobcalvert.com/teaching/math637fall2026/>

INSTRUCTOR Jacob Calvert (calvert@udel.edu)

OFFICE HOURS MW 2:45pm–3:45pm, Ewing Hall 509

Announcements & Reminders

- Quiz 2, on k -means and Lloyd's algorithm, will be Friday
- A worksheet (ungraded) for Lecture 5 is now available on Canvas
- Please complete the **anonymous** early semester feedback form linked on Canvas

Plan for today

- **Recap: fitting one Gaussian** 6 min
- Fitting a mixture of k Gaussians 10 min
- Interlude: responsibilities 8 min
- Back to fitting k Gaussians 6 min
- The EM algorithm 15 min
- Bivariate and multivariate Gaussians 10 min
- Choosing k 5 min

Recall: Gaussian mixture modeling

Problem (Gaussian mixture modeling in 1D)

Given points $x_1, \dots, x_n \in \mathbb{R}$ and a number of components k , find weights $\pi_1, \dots, \pi_k$, means $\mu_1, \dots, \mu_k$, and variances $\sigma_1^2, \dots, \sigma_k^2$ such that

$$p(x) = \sum_{j=1}^k \pi_j \mathcal{N}(x \mid \mu_j, \sigma_j^2)$$

maximizes the likelihood.

Note: By convention, $\theta = (\pi_1, \dots, \pi_k, \mu_1, \dots, \mu_k, \sigma_1^2, \dots, \sigma_k^2)$ denotes the entire list of parameters, and the likelihood and log-likelihood are

$$L(\theta) = \prod_{i=1}^n p_{\theta}(x_i), \quad \ell(\theta) = \log L(\theta).$$

Recall: Fitting **one** Gaussian

Problem (Gaussian mixture modeling in 1D)

Given points $x_1, \dots, x_n \in \mathbb{R}$, find a mean μ and variance σ^2 such that

$$p_{\theta}(x) = \mathcal{N}(x \mid \mu, \sigma^2)$$

maximizes the likelihood $L(\theta) = \prod_{i=1}^n p_{\theta}(x_i)$.

Recall: Fitting **one** Gaussian

Problem (Gaussian mixture modeling in 1D)

Given points $x_1, \dots, x_n \in \mathbb{R}$, find a mean μ and variance σ^2 such that

$$p(x) = \mathcal{N}(x \mid \mu, \sigma^2)$$

maximizes the **log-likelihood** $\ell(\theta) = \sum_{i=1}^n \log p_{\theta}(x_i)$.

Recall: The case $k = 1$

- With $k = 1$, $p_{\theta}(x) = \mathcal{N}(x \mid \mu, \sigma^2)$, so

$$\begin{aligned}\ell(\theta) &= \sum_{i=1}^n \log \mathcal{N}(x_i \mid \mu, \sigma^2) \\ &= \sum_{i=1}^n \log \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left(-\frac{(x_i - \mu)^2}{2\sigma^2} \right) \\ &= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2.\end{aligned}$$

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- Setting the derivatives to zero gave

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

Recall: Fitting **one** Gaussian

Problem (Gaussian mixture modeling in 1D)

Given points $x_1, \dots, x_n \in \mathbb{R}$, find a mean μ and variance σ^2 such that

$$p_{\theta}(x) = \mathcal{N}(x \mid \mu, \sigma^2)$$

maximizes the likelihood $L(\theta) = \prod_{i=1}^n p_{\theta}(x_i)$.

Idea: To fit a single Gaussian in 1D, set

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i \quad \text{and} \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

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Exercise: Find the log-likelihood for general k

For the Gaussian mixture model with k components, the density is

$$p_{\theta}(x) = \sum_{j=1}^k \pi_j \mathcal{N}(x \mid \mu_j, \sigma_j^2).$$

What is the log-likelihood $\ell(\theta)$ given examples $x_1, \dots, x_n$?

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$$\ell(\theta) = \sum_{i=1}^n \log \left(\sum_{j=1}^k \pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2) \right).$$

Key idea: There is now a sum inside the logarithm. This means that we will not be able to obtain explicit formulas for the parameters.

Differentiating the log-likelihood with respect to μ_j

- By the chain rule,

$$\frac{\partial \ell}{\partial \mu_j} = \sum_{i=1}^n \frac{\pi_j \frac{\partial}{\partial \mu_j} \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}{\sum_{l=1}^k \pi_l \mathcal{N}(x_i \mid \mu_l, \sigma_l^2)}.$$

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- Note that

$$\frac{\partial}{\partial \mu} \mathcal{N}(x \mid \mu, \sigma^2) = \mathcal{N}(x \mid \mu, \sigma^2) (x - \mu) / \sigma^2.$$

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- Note that

$$\frac{\partial}{\partial \mu} \mathcal{N}(x | \mu, \sigma^2) = \mathcal{N}(x | \mu, \sigma^2) (x - \mu) / \sigma^2.$$

- Combining gives

$$\frac{\partial \ell}{\partial \mu_j} = \sum_{i=1}^n \frac{\pi_j \mathcal{N}(x_i | \mu_j, \sigma_j^2)}{\underbrace{\sum_{l=1}^k \pi_l \mathcal{N}(x_i | \mu_l, \sigma_l^2)}_{r_{ij}}} \cdot \frac{x_i - \mu_j}{\sigma_j^2}.$$

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Definition: Responsibility

Definition

The **responsibility** that component j takes for x is

$$r_j(x) = \frac{\pi_j \mathcal{N}(x \mid \mu_j, \sigma_j^2)}{\sum_{l=1}^k \pi_l \mathcal{N}(x \mid \mu_l, \sigma_l^2)}, \quad r_{ij} = r_j(x_i).$$

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If $Z \sim \text{Categorical}(\pi_1, \dots, \pi_k)$ and $X \mid Z = j \sim \mathcal{N}(\mu_j, \sigma_j^2)$, then by **Bayes' rule**

$$\mathbb{P}(Z = j \mid X = x) = \frac{\mathbb{P}(Z = j) p(x \mid Z = j)}{p(x)} = \frac{\pi_j \mathcal{N}(x \mid \mu_j, \sigma_j^2)}{p(x)} = r_j(x).$$

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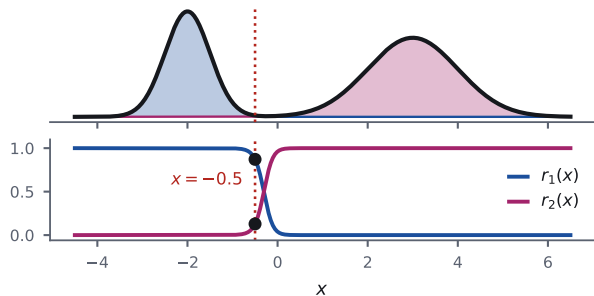
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Key idea: $r_j(x)$ is the probability that x came from component j !

Example: $0.4 \mathcal{N}(-2, 0.5^2) + 0.6 \mathcal{N}(3, 1)$

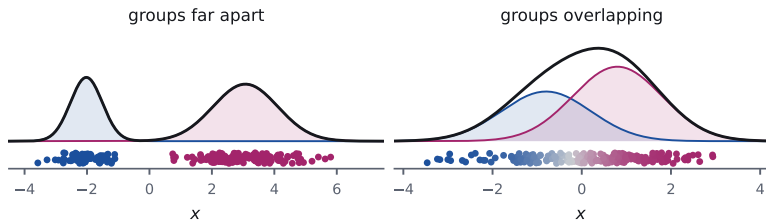


x	$r_1(x)$	$r_2(x)$
-2.0	1.000	0.000
-0.5	0.871	0.129
-0.3	0.500	0.500
0.0	0.039	0.961

$r_1(x) + r_2(x) = 1$ always.

Key idea: Instead of “ x belongs to component 1” we can say “ x came from component 1 with probability 0.871.”

How “soft” is soft assignment?



Colored by r_1 : 0 through grey to 1.

Watch out: On the left, **not one** of the 200 points has r_1 between 0.01 and 0.99; on the right, most lie between 0.05 and 0.95. The softness of an assignment is a property of the *overlap*, not the method.

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Setting the derivatives to zero

Setting

$$\frac{\partial \ell}{\partial \mu_j} = 0, \quad \frac{\partial \ell}{\partial \sigma_j^2} = 0, \quad \text{and} \quad \frac{\partial \ell}{\partial \pi_j} = 0 \quad \text{subject to} \quad \sum_j \pi_j = 1,$$

gives

$$\mu_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} x_i, \quad \sigma_j^2 = \frac{1}{n_j} \sum_{i=1}^n r_{ij} (x_i - \mu_j)^2, \quad \pi_j = \frac{n_j}{n},$$

where $n_j = \sum_{i=1}^n r_{ij}$ is the effective number of points for which component j is responsible, and $n_1 + \dots + n_k = n$.

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where $n_j = \sum_{i=1}^n r_{ij}$ is the effective number of points for which component j is responsible, and $n_1 + \dots + n_k = n$.

Watch out: Every r_{ij} depends on every π_l , μ_l and σ_l^2 , so these are not closed-form expressions for the parameters!

Key observation: We have seen these formulas

For $k = 1$ we found

$$\mu = \frac{1}{n} \sum_{i=1}^n x_i, \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2.$$

For general k we have just found

$$\mu_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} x_i, \quad \sigma_j^2 = \frac{1}{n_j} \sum_{i=1}^n r_{ij} (x_i - \mu_j)^2.$$

Key idea: The same two formulas, with each point **weighted** by how much component j is responsible for it.

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Algorithm: Expectation-maximization

Definition (EM for a Gaussian mixture)

Initialize π_j, μ_j, σ_j^2 . Until the log-likelihood stops changing, repeat:

1. **E step.** With the parameters fixed, compute

$$r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \sigma_j^2)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \sigma_l^2)}.$$

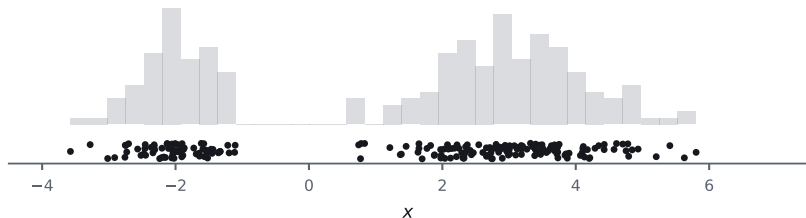
2. **M step.** With the r_{ij} fixed, set

$$\pi_j = \frac{n_j}{n} \quad \text{and} \quad \mu_j = \frac{1}{n_j} \sum_i r_{ij} x_i, \quad \text{then} \quad \sigma_j^2 = \frac{1}{n_j} \sum_i r_{ij} (x_i - \mu_j)^2.$$

Important: Update μ_j before σ_j^2 , and use the new μ_j in the second sum!

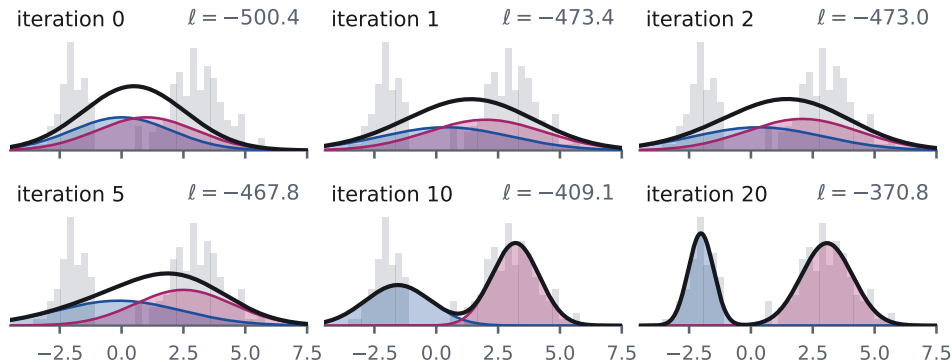
Recall: Example from lecture 6

Now suppose you are given $n = 200$ points.



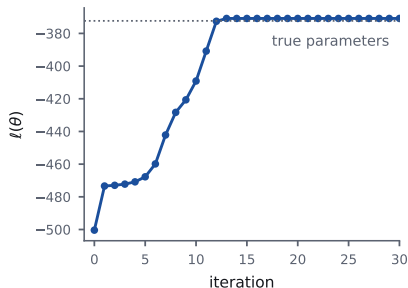
Model as a mixture of two Gaussians. But which two?

Example: EM on the same 200 points



Started from $\pi = (0.5, 0.5)^T$, $\mu = (0, 1)^T$, $\sigma = (2, 2)^T$.

Fact: $\ell(\theta)$ never decreases under EM iteration



Here ℓ climbs from -500.4 to -370.8 , ending at

$$\pi = (0.355, 0.645)^T, \quad \mu = (-2.02, 3.07)^T, \quad \text{and} \quad \sigma = (0.51, 1.04)^T.$$

Compare this to the true model:

$$0.4 \mathcal{N}(-2, 0.5^2) + 0.6 \mathcal{N}(3, 1).$$

Warning 1: Where it stops depends on where it starts

- Like the k -means objective J , the log-likelihood ℓ is **not concave**, so EM is not guaranteed to find the global maximizer.

Recall: Lloyd's algorithm had the same properties, and we handled them the same way.

Warning 1: Where it stops depends on where it starts

- Like the k -means objective J , the log-likelihood ℓ is **not concave**, so EM is not guaranteed to find the global maximizer.
 - With $k = 3$ on our 200 points, 200 random starts gave 100 distinct values of ℓ at convergence.

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- Technically, relabelling the components leaves ℓ unchanged, so every maximum comes with $k!$ copies of itself.

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 - With $k = 3$ on our 200 points, 200 random starts gave 100 distinct values of ℓ at convergence.
- Technically, relabelling the components leaves ℓ unchanged, so every maximum comes with $k!$ copies of itself.
- In practice: use several random starts, keep the best ℓ .

Recall: Lloyd's algorithm had the same properties, and we handled them the same way.

Is there a better way to initialize?

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Yes, use k -means (I'm not joking)!

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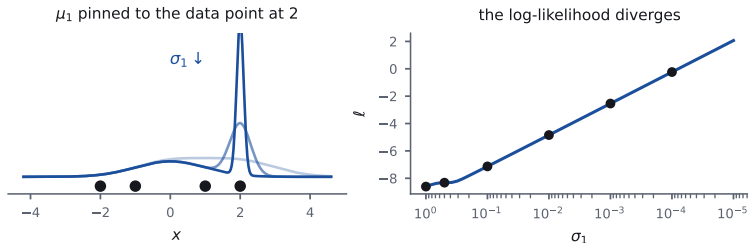
Yes, use k -means (I'm not joking)!

Specifically, to initialize $(\pi_j, \mu_j, \sigma_j^2)_{j=1}^k$:

1. Run k -means++ clustering
2. Set π_j to be the fraction of points assigned to cluster j and μ_j to be the centroids of these points
3. Then, set σ_j^2 to be the empirical variance of the points around μ_j

Warning 2: The likelihood is unbounded

Put μ_1 on a data point and let $\sigma_1 \rightarrow 0$.



Here $\{-2, -1, 1, 2\}$, with $\pi = (\frac{1}{2}, \frac{1}{2})^\top$, $\mu = (2, 0)^\top$ and $\sigma_2 = 1$.

Watch out: To avoid “variance collapse,” either use many more examples than components (i.e., $n \gg k$) or put a lower bound on the variances.

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Recall: The bivariate Gaussian

For $x \in \mathbb{R}^2$, a mean $\mu \in \mathbb{R}^2$ and a symmetric positive definite $\Sigma \in \mathbb{R}^{2 \times 2}$,

$$\mathcal{N}(x \mid \mu, \Sigma) = \frac{1}{2\pi |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right),$$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} \sigma_1^2 & \rho \sigma_1 \sigma_2 \\ \rho \sigma_1 \sigma_2 & \sigma_2^2 \end{pmatrix}.$$

- the diagonal of Σ holds the variances of x_1 and x_2

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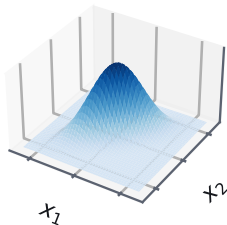
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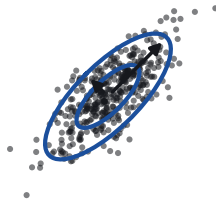
- the diagonal of Σ holds the variances of x_1 and x_2
- the off-diagonal holds their covariance; $\rho \in (-1, 1)$ is the correlation

Example: What they look like

the density



seen from above, with a sample



Recall: Here $\mu = 0$ and $\Sigma = \begin{pmatrix} 1 & 0.75 \\ 0.75 & 1 \end{pmatrix}$. The contours are ellipses, with axes the eigenvectors of Σ and half-lengths $\sqrt{\lambda_1} = 1.32$, $\sqrt{\lambda_2} = 0.5$.

Algorithm: EM in two dimensions

Definition (EM for a mixture of bivariate Gaussians)

Initialize π_j, μ_j, Σ_j . Until ℓ stops changing, repeat:

1. **E step.** $r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)}$, and $n_j = \sum_i r_{ij}$.

2. **M step.** $\pi_j = \frac{n_j}{n}$, $\mu_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} x_i$,

$$\Sigma_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} (x_i - \mu_j)(x_i - \mu_j)^\top.$$

Note: The π_j and μ_j formulas don't change; they just take vector values now. Instead of σ_j^2 , we have the matrix Σ_j .

What is Σ_j ? A weighted covariance

For $x, \mu \in \mathbb{R}^2$ the outer product is the 2×2 matrix

$$(x - \mu)(x - \mu)^T = \begin{pmatrix} (x_1 - \mu_1)^2 & (x_1 - \mu_1)(x_2 - \mu_2) \\ (x_1 - \mu_1)(x_2 - \mu_2) & (x_2 - \mu_2)^2 \end{pmatrix}.$$

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Averaging it with weights r_{ij} therefore puts the weighted variance of x_1 in the top-left, the weighted variance of x_2 in the bottom-right, and the weighted covariance off the diagonal.

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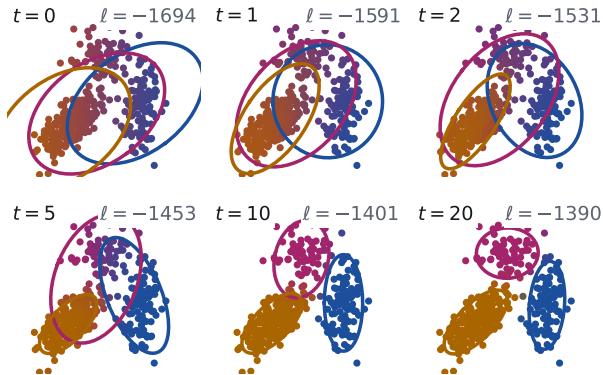
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Key idea: Σ_j is the sample covariance matrix of the data, with point i counted r_{ij} times. In one dimension it was the sample variance, counted the same way.

Example: EM on 400 points in $\mathbb{R}^2$



Each point is colored by mixing the three colors in proportion to

$$r_{i1}, r_{i2}, r_{i3}.$$

What about $\mathbb{R}^p$? Nothing changes

The same two steps, with $x_i, \mu_j \in \mathbb{R}^p$ and $\Sigma_j \in \mathbb{R}^{p \times p}$:

$$r_{ij} = \frac{\pi_j \mathcal{N}(x_i \mid \mu_j, \Sigma_j)}{\sum_l \pi_l \mathcal{N}(x_i \mid \mu_l, \Sigma_l)}, \quad n_j = \sum_{i=1}^n r_{ij},$$

$$\pi_j = \frac{n_j}{n}, \quad \mu_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} x_i, \quad \Sigma_j = \frac{1}{n_j} \sum_{i=1}^n r_{ij} (x_i - \mu_j)(x_i - \mu_j)^\top.$$

Idea: Generalizing variances to covariance matrices enables us to find ellipsoidal clusters instead of only spherical ones.

Compare: Lloyd's algorithm and EM

	Lloyd's algorithm	EM for a mixture
initialize	$\mu_1, \dots, \mu_k$	π_j, μ_j, Σ_j
step 1	nearest center wins	$r_{ij} = \mathbb{P}(Z = j \mid x_i)$
step 2	center $\leftarrow$ mean of its points	$\mu_j \leftarrow r$ -weighted mean
guarantee	J never increases	ℓ never decreases
output	one label per point	k probabilities per point
start matters	yes	yes

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A question: How many components?

- A k -component mixture contains every $(k - 1)$ -component mixture: set $\pi_k = 0$. So $\ell(\theta)$ can only go **up** as k grows.

Recall: J did the same thing for k -means, and we had nothing to set against it. Now we do.

A question: How many components?

- A k -component mixture contains every $(k - 1)$ -component mixture: set $\pi_k = 0$. So $\ell(\theta)$ can only go **up** as k grows.
- “Choose k to maximize the likelihood” therefore answers $k = n$.

Recall: J did the same thing for k -means, and we had nothing to set against it. Now we do.

Definition: Bayesian information criterion

Definition (BIC)

For parameters $\hat{\theta}$ fitted by maximum likelihood with m free parameters on n points,

$$\text{BIC} = -2 \ell(\hat{\theta}) + m \log n.$$

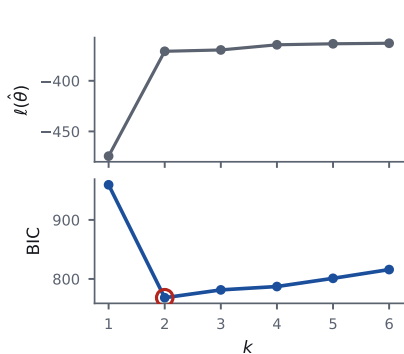
Idea: Among candidate models, choose the one that **minimizes** the BIC.

For a mixture of k Gaussians in $\mathbb{R}^p$ with full covariances,

$$m = \underbrace{k-1}_{\text{weights}} + \underbrace{kp}_{\text{means}} + \underbrace{kp(p+1)/2}_{\text{covariances}},$$

which is $3k - 1$ when $p = 1$, and 17 when $p = 2$ and $k = 3$.

Example: Choosing k for our 200 points



k	$\ell(\hat{\theta})$	m	BIC
1	-474.4	2	959.5
2	-370.8	5	768.2
3	-369.5	8	781.4
4	-364.4	11	787.1
5	-363.4	14	801.0
6	-362.9	17	815.9

Watch out: BIC only ranks the models you hand it. If none resembles the truth, its minimizer is the least bad of a bad list.